

```
[1]: # Create a list of 5 student marks
marks = [78, 85, 67, 92, 74]

print("Original List:", marks)

# append() - add a new mark
marks.append(88)

# insert() - insert a mark at a specific position
marks.insert(2, 80)

# remove() - remove a mark
marks.remove(67)

# sort() - sort the list in ascending order
marks.sort()

# Print updated list
print("Updated List:", marks)
```

Original List: [78, 85, 67, 92, 74]
Updated List: [74, 78, 80, 85, 88, 92]

```
[2]: # Create a tuple with 6 city names
cities = ("Delhi", "Mumbai", "Chennai", "Bangalore", "Hyderabad", "Kolkata")

# Print first city
print("First city:", cities[0])

# Print last city
print("Last city:", cities[-1])

# Count total elements
print("Total elements:", len(cities))

# Check whether Delhi exists or not
if "Delhi" in cities:
    print("Delhi exists in the tuple")
else:
    print("Delhi does not exist in the tuple")
```

First city: Delhi
Last city: Kolkata
Total elements: 6
Delhi exists in the tuple

```
[3]: # Create two sets
A = {1, 2, 3, 4, 5}
B = {4, 5, 6, 7}

# Union
print("Union:", A.union(B))

# Intersection
print("Intersection:", A.intersection(B))

# Difference
print("Difference (A - B):", A.difference(B))

# Symmetric Difference
print("Symmetric Difference:", A.symmetric_difference(B))
```

```
Union: {1, 2, 3, 4, 5, 6, 7}
Intersection: {4, 5}
Difference (A - B): {1, 2, 3}
Symmetric Difference: {1, 2, 3, 6, 7}
```

```
[4]: # Create a student dictionary
student = {
    "name": "Rahul",
    "roll": 101,
    "branch": "Computer Science",
    "marks": 85
}

# Update marks
student["marks"] = 92

# Print keys separately
print("Keys:")
for key in student.keys():
    print(key)

# Print values separately
print("\nValues:")
for value in student.values():
    print(value)
```

```
Keys:
name
roll
branch
marks
```

```
Values:
Rahul
101
Computer Science
92
```

--

```
[5]: # Store item names in a list
items = ["Pen", "Book", "Bag", "Bottle"]

# Store prices in another list
prices = [20, 150, 500, 80]

# Calculate total bill
total_bill = sum(prices)

# Find highest price
highest_price = max(prices)

# Find lowest price
lowest_price = min(prices)

# Display results
print("Items:", items)
print("Prices:", prices)
print("Total Bill:", total_bill)
print("Highest Price:", highest_price)
print("Lowest Price:", lowest_price)
```

```
Items: ['Pen', 'Book', 'Bag', 'Bottle']
Prices: [20, 150, 500, 80]
Total Bill: 750
Highest Price: 500
Lowest Price: 20
```

```
[6]: # Function for addition
def addition(a, b):
    return a + b

# Function for subtraction
def subtraction(a, b):
    return a - b

# Function for multiplication
def multiplication(a, b):
    return a * b

# Function for division
def division(a, b):
    if b != 0:
        return a / b
    else:
        return "Division by zero is not possible"

# Take input from user
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

# Display results
print("Addition:", addition(num1, num2))
print("Subtraction:", subtraction(num1, num2))
print("Multiplication:", multiplication(num1, num2))
print("Division:", division(num1, num2))

Enter first number: 1
Enter second number: 2
Addition: 3.0
Subtraction: -1.0
Multiplication: 2.0
Division: 0.5
```

```
[7]: # Function to check even or odd
def check_even_odd(n):
    if n % 2 == 0:
        return "Even"
    else:
        return "Odd"

# Take input from user
num = int(input("Enter a number: "))

# Display result
print("The number is:", check_even_odd(num))

Enter a number: 5
The number is: Odd
```

The number is: Odd

```
[8]: # Function to find the largest among three numbers
def largest(a, b, c):
    if a >= b and a >= c:
        return a
    elif b >= a and b >= c:
        return b
    else:
        return c

# Take input from user
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
num3 = int(input("Enter third number: "))

# Display result
print("Largest number is:", largest(num1, num2, num3))
```

Enter first number: 2
Enter second number: 5
Enter third number: 6
Largest number is: 6

```
[9]: # Function to calculate factorial
def factorial(n):
    fact = 1

    for i in range(1, n + 1):
        fact = fact * i

    return fact

# Take input from user
num = int(input("Enter a number: "))

# Display result
print("Factorial is:", factorial(num))
```

Enter a number: 2
Factorial is: 2

[10]: *# Function to check whether a number is prime*

```
def is_prime(n):  
    if n <= 1:  
        return False  
  
    for i in range(2, n):  
        if n % i == 0:  
            return False  
  
    return True  
  
# Take input from user  
num = int(input("Enter a number: "))  
  
# Display result  
if is_prime(num):  
    print("True")  
else:  
    print("False")
```

Enter a number: 2

True

[11]: *# List to store student names*

```
students = []  
  
# Function to add student  
def add_student(name):  
    students.append(name)  
    print(name, "added successfully")  
  
# Function to remove student  
def remove_student(name):  
    if name in students:  
        students.remove(name)  
        print(name, "removed successfully")  
    else:  
        print("Student not found")  
  
# Function to display students  
def display_students():  
    print("Student List:")  
    for student in students:  
        print(student)  
  
# Example  
add_student("Rahul")  
add_student("Anjali")  
remove_student("Rahul")  
display_students()
```

Rahul added successfully
Anjali added successfully
Rahul removed successfully
Student List:
Anjali


```
[12]: # Dictionary to store employee names and salaries
employees = {}

# Function to add employee
def add_employee(name, salary):
    employees[name] = salary
    print(name, "added successfully")

# Function to update salary
def update_salary(name, salary):
    if name in employees:
        employees[name] = salary
        print("Salary updated")
    else:
        print("Employee not found")

# Function to search employee
def search_employee(name):
    if name in employees:
        print(name, "Salary:", employees[name])
    else:
        print("Employee not found")

# Example
add_employee("John", 30000)
add_employee("David", 45000)

update_salary("John", 35000)

search_employee("John")

John added successfully
David added successfully
Salary updated
John Salary: 35000
```

```
[3]: # Take 10 numbers from user
numbers = []

for i in range(10):
    num = int(input("Enter number: "))
    numbers.append(num)

# Convert list to set
unique_numbers = set(numbers)

# Print unique count
print("Unique numbers:", unique_numbers)
print("Count of unique numbers:", len(unique_numbers))
```

```
Enter number: 1
Enter number: 2
Enter number: 2
Enter number: 3
Enter number: 4
Enter number: 5
Enter number: 6
Enter number: 5
Enter number: 4
Enter number: 5
Unique numbers: {1, 2, 3, 4, 5, 6}
Count of unique numbers: 6
```

```
[4]: # Store ages in tuple
ages = (16, 18, 22, 14, 30, 17)

# Function to check voting eligibility
def check_voting(age):
    if age >= 18:
        return "Eligible for voting"
    else:
        return "Not eligible"

# Display results
for age in ages:
    print("Age:", age, "-", check_voting(age))
```

```
Age: 16 - Not eligible
Age: 18 - Eligible for voting
Age: 22 - Eligible for voting
Age: 14 - Not eligible
Age: 30 - Eligible for voting
Age: 17 - Not eligible
```

```
[5]: # Take sentence from user
sentence = input("Enter a sentence: ")

# Split sentence into words
words = sentence.split()

# Create dictionary
frequency = {}

for word in words:
    if word in frequency:
        frequency[word] += 1
    else:
        frequency[word] = 1

# Display result
print(frequency)
```

```
Enter a sentence: python anaconda
{'python': 1, 'anaconda': 1}
```



```

[6]: # List to store books
books = []

# Functions
def add_book():
    book = input("Enter book name: ")
    books.append(book)
    print("Book added successfully")

def remove_book():
    book = input("Enter book name to remove: ")
    if book in books:
        books.remove(book)
        print("Book removed successfully")
    else:
        print("Book not found")

def search_book():
    book = input("Enter book name to search: ")
    if book in books:
        print("Book found")
    else:
        print("Book not found")

def display_books():
    print("Books List:")
    for book in books:
        print(book)

# Menu
while True:
    print("\n1.Add Book")
    print("2.Remove Book")
    print("3.Search Book")
    print("4.Display Books")
    print("5.Exit")

    choice = int(input("Enter choice: "))

    if choice == 1:
        add_book()
    elif choice == 2:
        remove_book()
    elif choice == 3:
        search_book()
    elif choice == 4:
        display_books()
    elif choice == 5:
        break
    else:
        print("Invalid choice")

```

```
1.Add Book
2.Remove Book
3.Search Book
4.Display Books
5.Exit
Enter choice: 1
Enter book name: python
Book added successfully
```

```
1.Add Book
2.Remove Book
3.Search Book
4.Display Books
5.Exit
Enter choice: 3
Enter book name to search: python
Book found
```

```
1.Add Book
2.Remove Book
3.Search Book
4.Display Books
5.Exit
Enter choice: 5
```

```
[7]: # Input marks of 5 students
marks = []

for i in range(5):
    mark = int(input("Enter marks: "))
    marks.append(mark)

# Functions
def average(marks):
    return sum(marks) / len(marks)

def highest(marks):
    return max(marks)

def lowest(marks):
    return min(marks)

def passed_students(marks):
    count = 0
    for mark in marks:
        if mark > 40:
            count += 1
    return count

# Display results
print("Average:", average(marks))
print("Highest:", highest(marks))
print("Lowest:", lowest(marks))
print("Passed Students:", passed_students(marks))
```

Enter marks: 530

Enter marks: 520

```

def lowest(marks):
    return min(marks)

def passed_students(marks):
    count = 0
    for mark in marks:
        if mark > 40:
            count += 1
    return count

# Display results
print("Average:", average(marks))
print("Highest:", highest(marks))
print("Lowest:", lowest(marks))
print("Passed Students:", passed_students(marks))

```

```

Enter marks: 530
Enter marks: 530
Enter marks: 530
Enter marks: 400
Enter marks: 233
Average: 444.6
Highest: 530
Lowest: 233
Passed Students: 5

```

```

[8]: # Function to check password strength
def check_password(password):

    has_digit = False
    has_upper = False
    has_lower = False

    for ch in password:
        if ch.isdigit():
            has_digit = True
        elif ch.isupper():
            has_upper = True
        elif ch.islower():
            has_lower = True

    if len(password) >= 8 and has_digit and has_upper and has_lower:
        print("Strong Password")
    else:
        print("Weak Password")

# Input password
password = input("Enter password: ")

# Call function
check_password(password)

```

```

Enter password: kusuma
Weak Password

```

```
[9]: # Dictionary to store contacts
contacts = {}

# Functions
def add_contact():
    name = input("Enter contact name: ")
    number = input("Enter phone number: ")
    contacts[name] = number
    print("Contact added successfully")

def search_contact():
    name = input("Enter contact name: ")

    if name in contacts:
        print("Phone Number:", contacts[name])
    else:
        print("Contact not found")

def delete_contact():
    name = input("Enter contact name: ")

    if name in contacts:
        del contacts[name]
        print("Contact deleted successfully")
    else:
        print("Contact not found")

def display_contacts():
    print("Contacts:")
    for name, number in contacts.items():
        print(name, ":", number)

# Menu
while True:
    print("\n1.Add Contact")
    print("2.Search Contact")
    print("3.Delete Contact")
    print("4.Display Contacts")
    print("5.Exit")

    choice = int(input("Enter choice: "))

    if choice == 1:
        add_contact()
    elif choice == 2:
        search_contact()
    elif choice == 3:
        delete_contact()
    elif choice == 4:
        display_contacts()
    elif choice == 5:
        break
    else:
        print("Invalid choice")
```

```
1.Add Contact
2.Search Contact
3.Delete Contact
4.Display Contacts
5.Exit
Enter choice: 1
Enter contact name: 3606140902
Enter phone number: sinchana
Contact added successfully
```

```
1.Add Contact
2.Search Contact
3.Delete Contact
4.Display Contacts
5.Exit
Enter choice: 4
Contacts:
3606140902 : sinchana
```

```
1 Add Contact
```

3606140902 : sinchana

```
1.Add Contact
2.Search Contact
3.Delete Contact
4.Display Contacts
5.Exit
Enter choice: 5
```

```
[10]: # Global balance variable
balance = 1000

# Deposit function
def deposit(amount):
    global balance
    balance += amount
    print("Amount deposited successfully")

# Withdraw function
def withdraw(amount):
    global balance

    if amount <= balance:
        balance -= amount
        print("Withdrawal successful")
    else:
        print("Insufficient balance")

# Check balance function
def check_balance():
    print("Current Balance:", balance)

# Example
deposit(500)
withdraw(300)
check_balance()
```

```
Amount deposited successfully
Withdrawal successful
Current Balance: 1200
```

```
[11]: # Given list
numbers = [10, 20, 30, 40, 50]

# Find maximum without max()
maximum = numbers[0]

for num in numbers:
    if num > maximum:
        maximum = num

# Find minimum without min()
minimum = numbers[0]

for num in numbers:
    if num < minimum:
        minimum = num

# Reverse list without reverse()
reversed_list = []

for i in range(len(numbers)-1, -1, -1):
    reversed_list.append(numbers[i])

# Display results
print("Maximum:", maximum)
print("Minimum:", minimum)
print("Reversed List:", reversed_list)
```

```
Maximum: 50
Minimum: 10
Reversed List: [50, 40, 30, 20, 10]
```